

Vladimir Strokov

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EDUCATION

Johns Hopkins University , Department of Physics and Astronomy Ph.D. , gravitational waves Advisor: Emanuele Berti; co-advisor: Giacomo Fragione (Northwestern)	Baltimore, USA 2020–2024
Lebedev Physical Institute , Astro Space Centre Candidate of Sciences , gravity theory. Advisor: V.N. Lukash	Moscow, Russia 2011
Moscow Institute of Physics and Technology M.Sc., <i>with excellence</i> / B.Sc., <i>with excellence</i>	Moscow, Russia 2007 / 2005

RESEARCH

West Virginia University , Center for Gravitational Waves and Cosmology GWAC Postdoctoral Fellow , LISA gravitational-wave science	Morgantown, USA 2024–present
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- LISA data analysis: the global fit, signal denoising with Hankel low-rank matrix approximation, source mixing (Space@Hopkins seed grant: proposal co-writer, collaborator)
- Galactic double white dwarfs: confusion foreground; co-coordinator of two LISA Consortium projects (approval pending)
- Black-hole shadows: image recovery and parameter inference with neural networks (NSF ACCESS allocation, PI)

Johns Hopkins University , Department of Physics and Astronomy Research Assistant	Baltimore, USA 2020–2024
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Intermediate-mass black holes and double white dwarfs with LISA

Lebedev Physical Institute , Astro Space Centre Lab Assistant / Research Fellow (on medical leave: 2014–2017)	Moscow, Russia 2005–2020
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- A neural network inferring black-hole parameters from partial interferometric images of the shadow
- GPU ray tracing for simulations of black-hole shadows

Federal Universities of Espírito Santo and Juiz de Fora Postdoctoral researcher : modified gravity; field theory in curved space–time	Brazil 2012–2014
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PROFESSIONAL MEMBERSHIPS

- LISA Consortium (member since 2020; **core member** since 2025)
- American Physical Society (since 2021)
- American Astronomical Society (since 2024)
- International Society on General Relativity and Gravitation (since 2022)

PROFESSIONAL SERVICE

- Referee for *Physical Review D*, *Classical and Quantum Gravity*, *European Physical Journal C*, *The Astrophysical Journal*, and *Astronomy & Astrophysics* (2017–present: 20 reports completed + 2 in progress)
- Session chair: APS April Meeting 2024, Sacramento; NANOGrav Fall Meeting 2025, Missoula
- Organizing committee, II José Plínio Baptista School of Cosmology, Brazil (2014)

TEACHING

- Sole instructor, graduate Classical Mechanics, *West Virginia University* (Spring 2027, scheduled)
- Sole instructor, graduate Advanced Electromagnetism 1, *West Virginia University* (Spring 2026)
- Designed and taught the intersession course “MCMC: a lever to move the world of data”, *Johns Hopkins University* (2023)
- Teaching assistant, graduate Electromagnetic Theory (2022–2024) and Quantum Mechanics I (2022), *Johns Hopkins University*
- 2023 Teaching Award, Department of Physics and Astronomy, *Johns Hopkins University*
- Physics bootcamp instructor and program tutor (2022–2023), Post-Baccalaureate Premedical Program, *Johns Hopkins University*
- Research mentor of a high-school student, project *Visualizing Gravity in GR* (2022)
- Undergraduate courses in cosmology, astronomy, and general relativity, *Moscow Institute of Physics and Technology* and *Federal University of Esp rito Santo* (2013–2019)

PUBLIC ENGAGEMENT

- Sonification of LISA signals, open source (github.com/CosmoVlad/gw-sounds)
- Invited seminar on gravitational waves for undergraduate students, PLASTRO REU program, *West Virginia University* (2026)
- Invited speaker on research careers in physics, BioComp 2023, a summer school for high-school students, *Johns Hopkins University* (2023)
- Trained to operate the 20-inch Morris W. Offit Telescope for public observing nights, Maryland Space Grant Observatory, *Johns Hopkins University* (2024)
- Russian translations of popular-science books: P. Halpern, *Collider* (Moscow: Eksmo, 2010); S. Weinberg, *The First Three Minutes* (Moscow: Eksmo, 2011)

TECHNICAL SKILLS

- Standard Python libraries: `numpy`, `scipy`, `matplotlib`, `astropy`
- Machine learning: `jax`, `tensorflow` (`keras`), `scikit-learn`
- AI-assisted development: Claude, Gemini, Antigravity
- High-performance computing: OpenMP, CUDA C/C++, `pycuda`, `slurm/launcher`
- General: `git`, $\text{\LaTeX}$

LANGUAGES

Russian: Native.

English/Portuguese: Proficient.

French: Advanced.

German: Reading Knowledge.

PRESENTATIONS

“Gravitational-wave source mixing in LISA” (talk), <i>WVU Center for Gravitational Waves and Cosmology Symposium</i> , Morgantown, WV	2026
“Galactic Double White Dwarfs: A Case of Cosmic Confusion” (invited seminar), <i>PLASTRO REU program</i> , <i>West Virginia University</i> , Morgantown, WV	2026
“Confusion foregrounds of DWD populations in the UCB Catalogs project” (poster), <i>16th International LISA Symposium</i> , College Park, MD	2026

“Hankel low-rank matrix approximation for time series denoising in gravitational-wave science” (talk), <i>APS Global Physics Summit 2026</i> , Denver, CO	2026
“Hankel low-rank matrix approximation for time series denoising” (talk), <i>NANOGrav Fall 2025 Meeting</i> , Missoula, MT	2025
“LISA double white dwarf binaries as Galactic accelerometers” (talk), <i>11th Gulf Coast Gravity Meeting</i> , University of Mississippi, Oxford, MS	2025
“LISA double white dwarf binaries as Galactic accelerometers” (contributed talk), <i>Ten Years to LISA Workshop</i> , NASA JPL, Pasadena, CA	2025
“Inferring a population of intermediate-mass black holes with LISA” (talk), <i>245th Meeting of the American Astronomical Society</i> , National Harbor, MD	2025
“Quasimonochromatic LISA sources in the Frequency Domain” (talk), <i>APS April Meeting 2024</i> , Sacramento, CA	2024
“Inferring a Population of Intermediate-Mass Black Holes with LISA” (poster), <i>APS April Meeting 2024</i> , Sacramento, CA	2024
“Quasimonochromatic LISA sources in the frequency domain” (talk), <i>6th Neighborhood Workshop on Astrophysics and Cosmology</i> , Penn State University, State College, PA	2024
“LISA Constraints on an Intermediate-Mass Black Hole in the Galactic Center” (talk), <i>MODEST-23: Star Clusters in the Post-Pandemic Era</i> , Evanston, IL	2023
“LISA Constraints on an Intermediate-Mass Black Hole in the Galactic Center” (talk), <i>APS April Meeting 2023</i> , Minneapolis, MN	2023
“Hunting for intermediate-mass black holes with LISA binary radial velocity measurements” (talk), <i>Intermediate-Mass Black Holes: New Science from Stellar Evolution to Cosmology</i> , San Juan, Puerto Rico	2022
“Hunting for intermediate-mass black holes with LISA binary radial velocity measurements” (virtual talk), <i>23rd International Conference on General Relativity and Gravitation (GR23)</i> , Liyang, China	2022
“Hunting for intermediate-mass black holes with LISA binary radial velocity measurements” (talk), <i>APS April Meeting 2022</i> , New York, NY	2022
“The orbital Lense–Thirring precession in a strong field” (talk), <i>10th Alexander Friedmann International Seminar on Gravitation and Cosmology</i> , St. Petersburg, Russia	2019
“Exoplanetary candidates to observations with Millimetron” (talk), <i>42nd COSPAR Scientific Assembly</i> , Pasadena, CA	2018
“The full cycle of simulating black hole shadows: from ray tracing to spectrum-specific images of an arbitrarily shaped screen” (poster), <i>42nd COSPAR Scientific Assembly</i> , Pasadena, CA	2018
“Nonstandard objects and theories of gravity” (talk), <i>XXXIII Conference on Topical Problems of Extragalactic Astronomy</i> , Pushchino, Russia	2016
“Numeric integration of geodesic equations in Schwarzschild’s field in different coordinate frames” (poster), <i>59th MIPT Scientific Conference</i> , Moscow, Russia	2016
“Black-and-white holes” (talk), <i>12th International Conference on Gravitation, Astrophysics and Cosmology (ICGAC-12)</i> , Moscow, Russia	2015
“Black-and-white hole as a space-time with integrable singularity” (talk), <i>9th Alexander Friedmann International Seminar on Gravitation and Cosmology</i> , Moscow, Russia	2015
“Phenomenological Approach in the Study of Singularity Problem” (contributed presentation), <i>26th Texas Symposium on Relativistic Astrophysics</i> , São Paulo, Brazil	2012